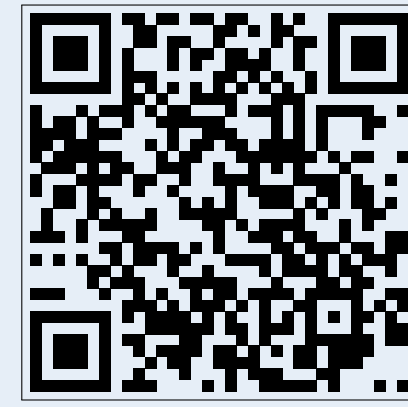




Abstract

This project addresses a foundational question in quantitative finance: when an options pricing model produces a theoretical fair value that diverges from the observed market premium, is that divergence large enough — and reliable enough — to justify a trade? A Cox–Ross–Rubinstein (CRR) American binomial option pricer is built in vectorized Python and validated against a live AMD option chain from Fidelity (April 2026). Its mispricing-edge signal is then cross-validated against an independent XGBoost ITM/OTM probability classifier trained on 101,535 historical AAPL contracts. The two layers agree strongly in the herding regime (Pearson correlation $r = 0.413$, $p = 2.4 \times 10^{-10}$) and not at all in the normal regime ($r \approx 0$). A regime-conditioned Kelly Criterion framework converts the cross-validated edge into risk-controlled capital allocation.

Keywords: CRR binomial pricing, options mispricing, XGBoost, Kelly Criterion, behavioral finance, EMH, herding regime.

Introduction & Motivation

Options markets aggregate beliefs into prices much like prediction markets do, but behavioral forces — herding around round strikes, implied-volatility overpricing driven by fear, and systematic skew in out-of-the-money puts — distort prices in predictable ways. The gap between a pricing model’s theoretical fair value and the observed market premium is potentially exploitable by a disciplined, model-driven strategy, but only if the divergence is large enough, reliable enough, and detectable by independent methods.

Research Questions

- Do CRR model prices predict whether an equity option expires in-the-money or out-of-the-money?
- Does cross-validation between CRR (math) and an ML classifier detect exploitable pricing *edge*?
- Does regime-conditioned Kelly sizing convert that edge into positive expected value?

Objectives

- Implement the CRR American binomial lattice (no dividends) in vectorized Python at $N = 100$ steps.
- Validate pricing accuracy against AMD market premiums within $\pm 5\%$ error.
- Train an XGBoost ITM/OTM classifier on engineered features *without* using IV as an input (independence from L1).
- Apply Kelly Criterion (full / half / quarter) for risk-controlled position sizing.

Monte Carlo Simulation

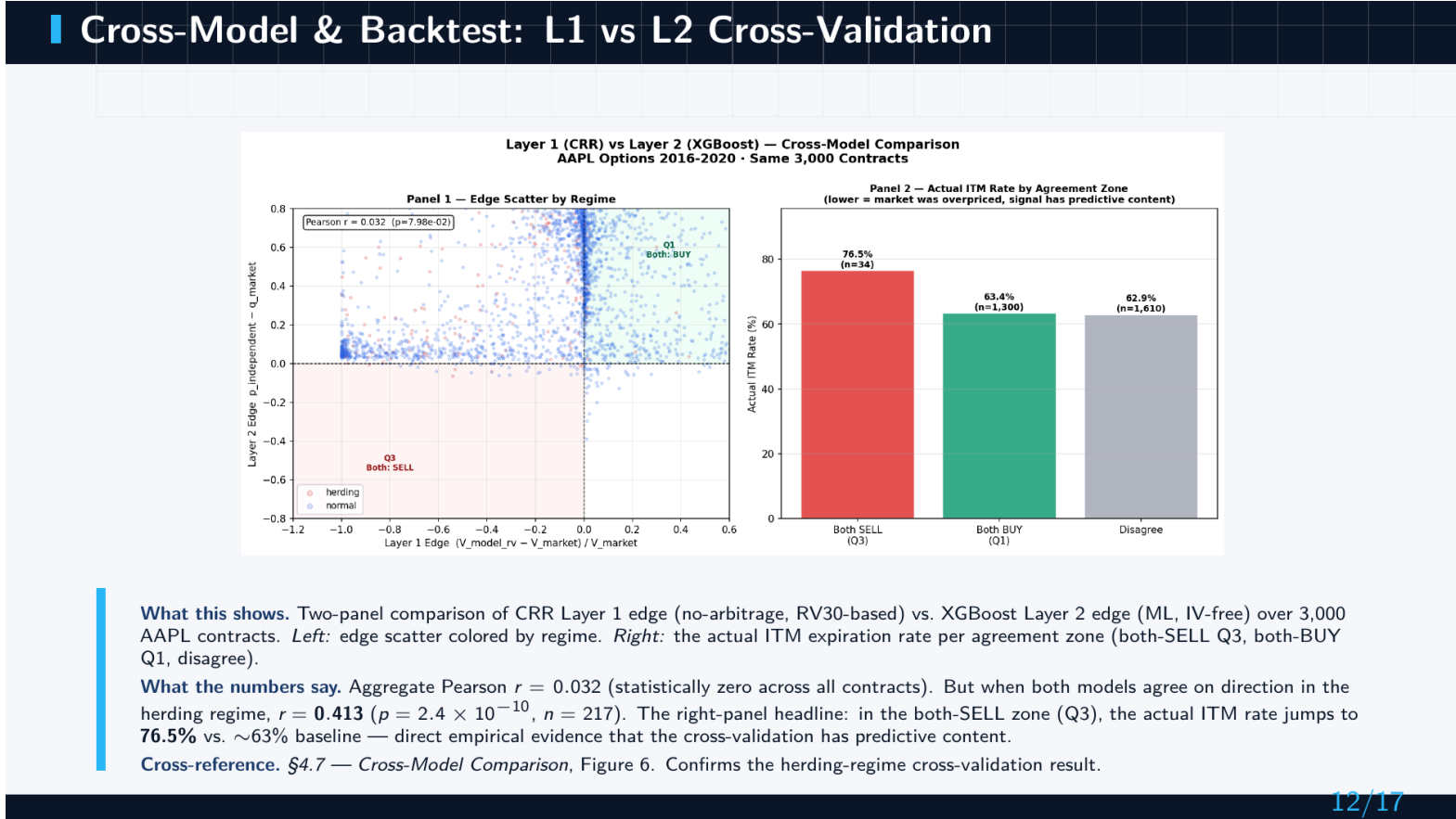
A Monte Carlo simulation validates that the Kelly Criterion converts the cross-validated edge into positive expected value under realistic trade-by-trade dynamics. One thousand trades are drawn from the empirical edge distribution with $p_{\text{win}} = 0.677$, and the cumulative profit-and-loss is tracked under full-, half-, and quarter-Kelly position sizing.



Reading the curves. Full Kelly (green) reaches $\sim \$12\text{M}$; half-Kelly (blue) $\$6\text{M}$; quarter-Kelly (orange) $\$3\text{M}$. All three variants share Sharpe = 5.93, confirming the Kelly fraction trades absolute return for drawdown control without changing risk-adjusted performance. The live pipeline runs half-Kelly in normal regimes and quarter-Kelly in herding regimes to keep the realized max drawdown below the 15% circuit-breaker threshold.

Exploratory Data Analysis

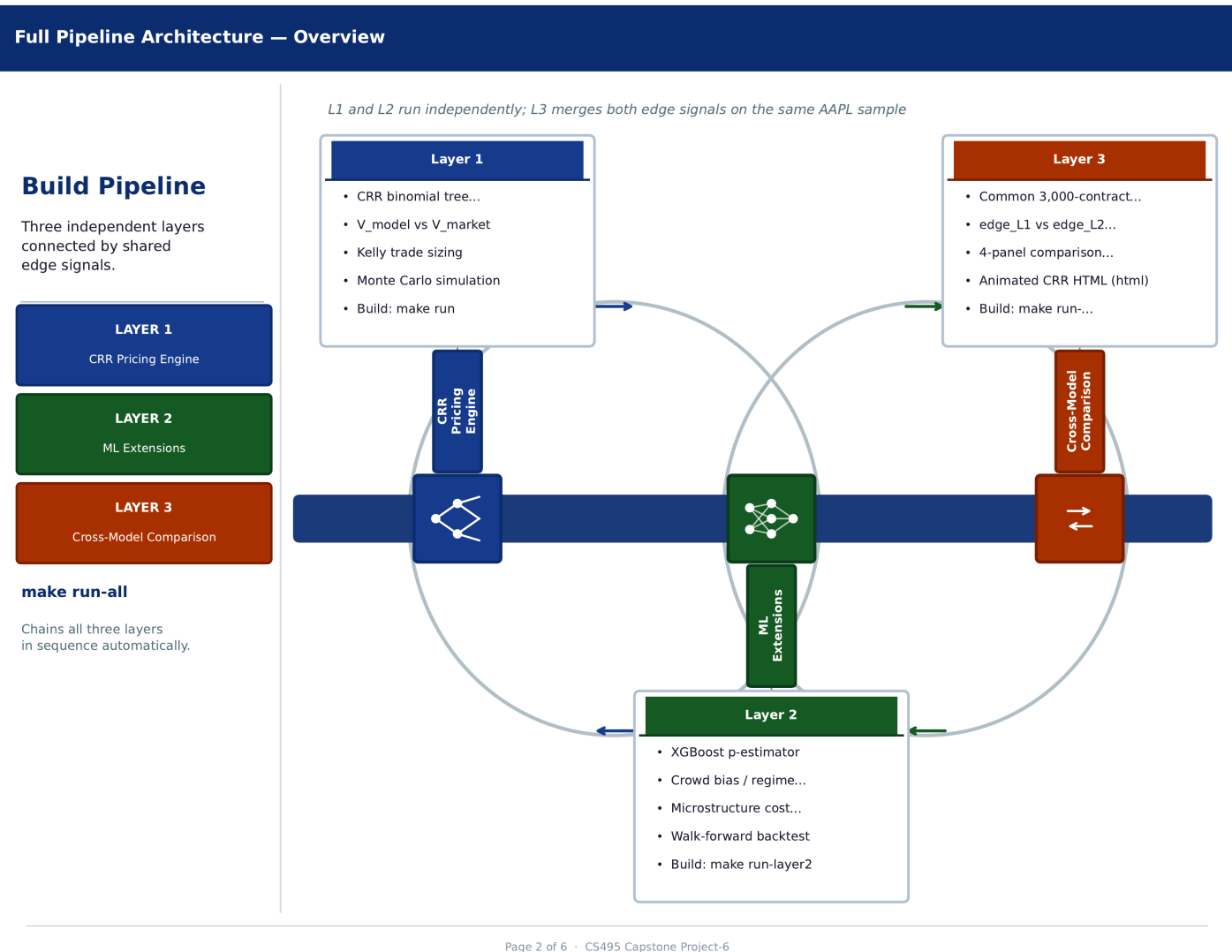
Cross-validating the two independent edge signals on 3,000 AAPL contracts reveals the central empirical finding of the project: the two models agree on direction precisely in the *herding* regime (red), not in the *normal* regime (blue).



Key pattern. *Left panel:* Layer-1 vs Layer-2 edge scatter colored by regime; the pooled Pearson correlation is $r = 0.032$ ($p = 0.080$), statistically zero across all contracts. *Right panel:* the actual ITM expiration rate per agreement zone — when both layers say SELL (Q3, $n = 34$), the realized ITM rate jumps to **76.5%** versus the $\sim 63\%$ baseline. The aggregate-vs-conditional inversion is a textbook Simpson’s paradox: r rises to **0.413** ($p = 2.4 \times 10^{-10}$, $n = 217$) once herding-regime contracts are isolated. This is the empirical fingerprint of *Samuelson macro-inefficiency* — markets are micro-efficient on average but exhibit aggregate bias under coordinated behavioral demand.

Methodology

A three-layer pipeline runs CRR and XGBoost independently, then merges their edge signals through a regime-conditioned Kelly position sizer.

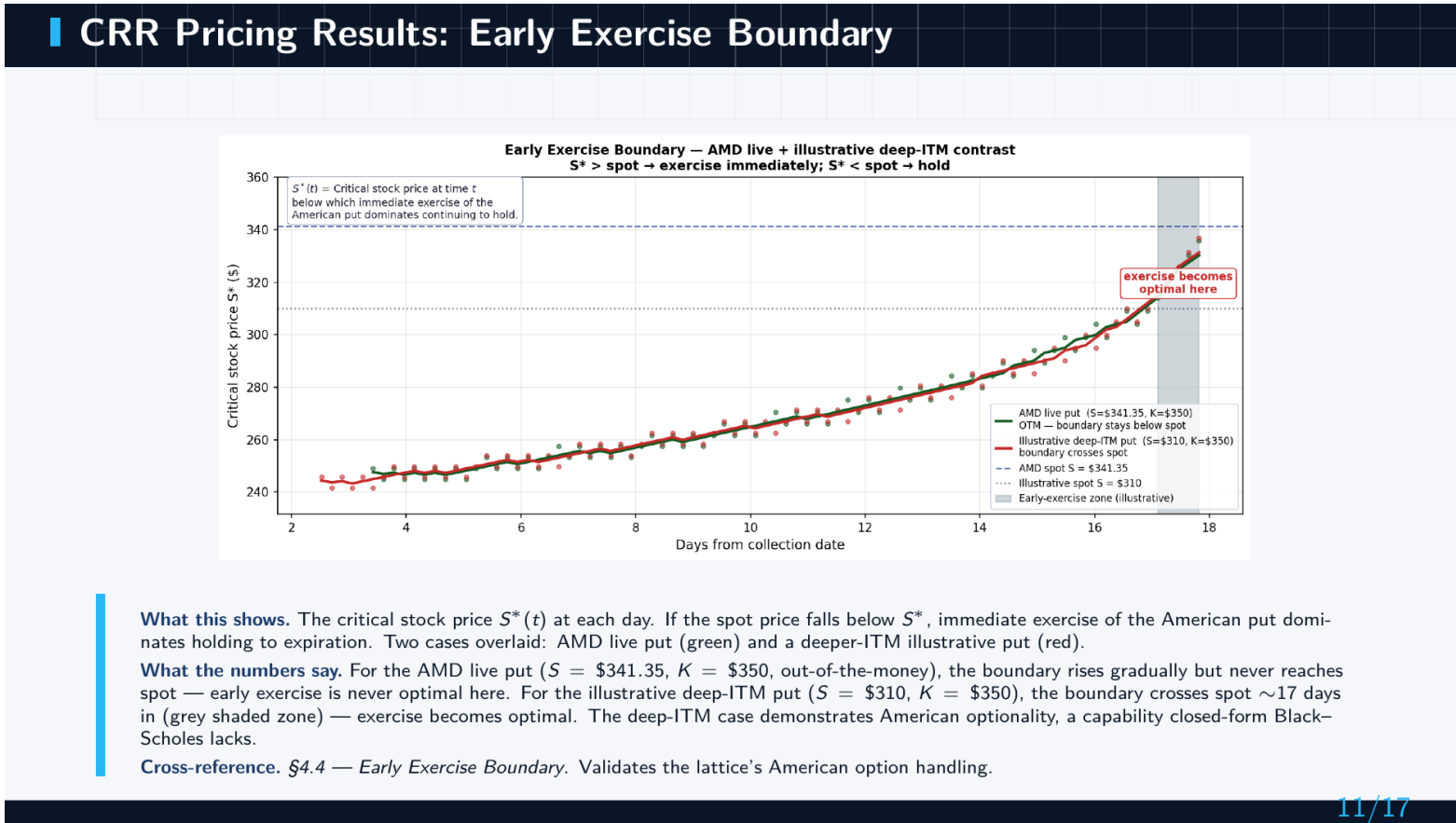


Techniques used

- Vectorized recombining lattice in NumPy ($N=50$ to 100 steps).
- Backward induction with American early-exercise check at each node.
- Greeks via centered finite differences.
- XGBoost trained on engineered features only (σ input is RV30, never IV — no leakage from L1).
- Platt scaling calibration; Brier-score validation.
- Walk-forward 60/20/20 split; zero look-ahead bias.

Early Exercise Boundary

The American-style optionality of the AMD put contracts is the principal reason the CRR binomial lattice is required: Black–Scholes omits early exercise. The boundary $S^*(t)$ is the critical stock price at each day t below which immediate exercise of the put dominates holding to expiration. The figure overlays two cases drawn from the lattice to isolate the regime in which the early-exercise feature actually triggers.

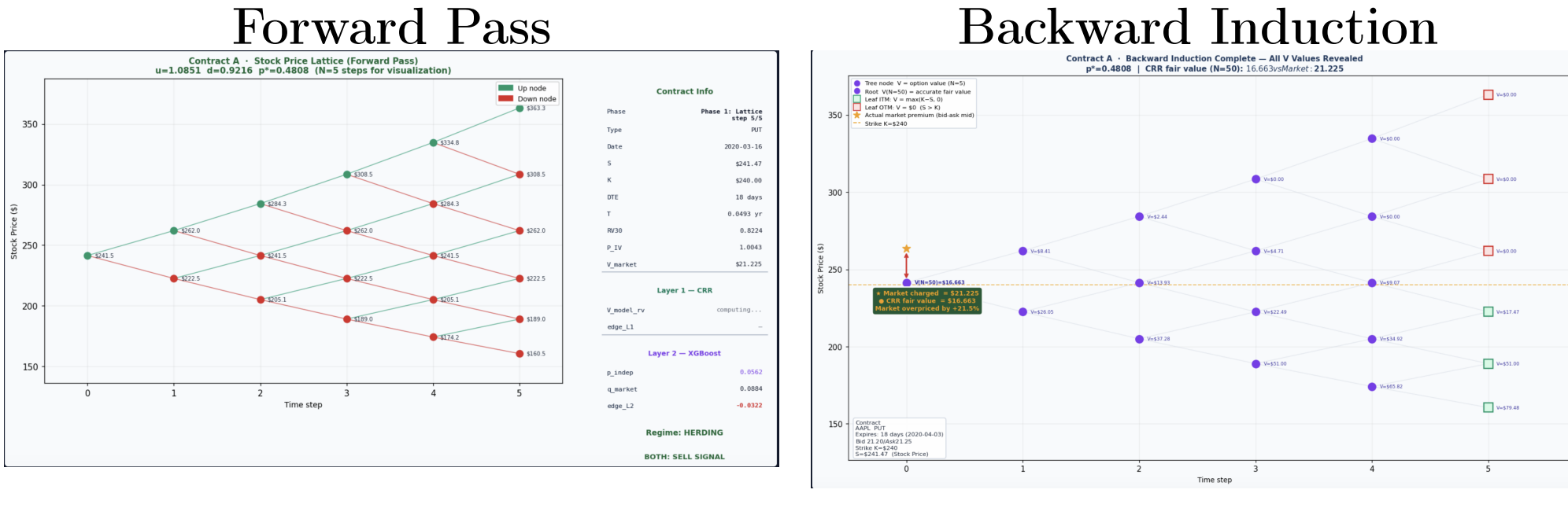


AMD live put (green). $S_0 = \$341.35$ above $K = \$350$ — the boundary rises gradually but never reaches spot, so the American premium exists in the model but never triggers as cash flow.

Deep-ITM put (red). $S_0 = \$310$ well below $K = \$350$ — the boundary crosses spot ~ 17 days in (grey zone), at which point early exercise becomes optimal. This isolates exactly the contracts a European-only pricer would mishandle.

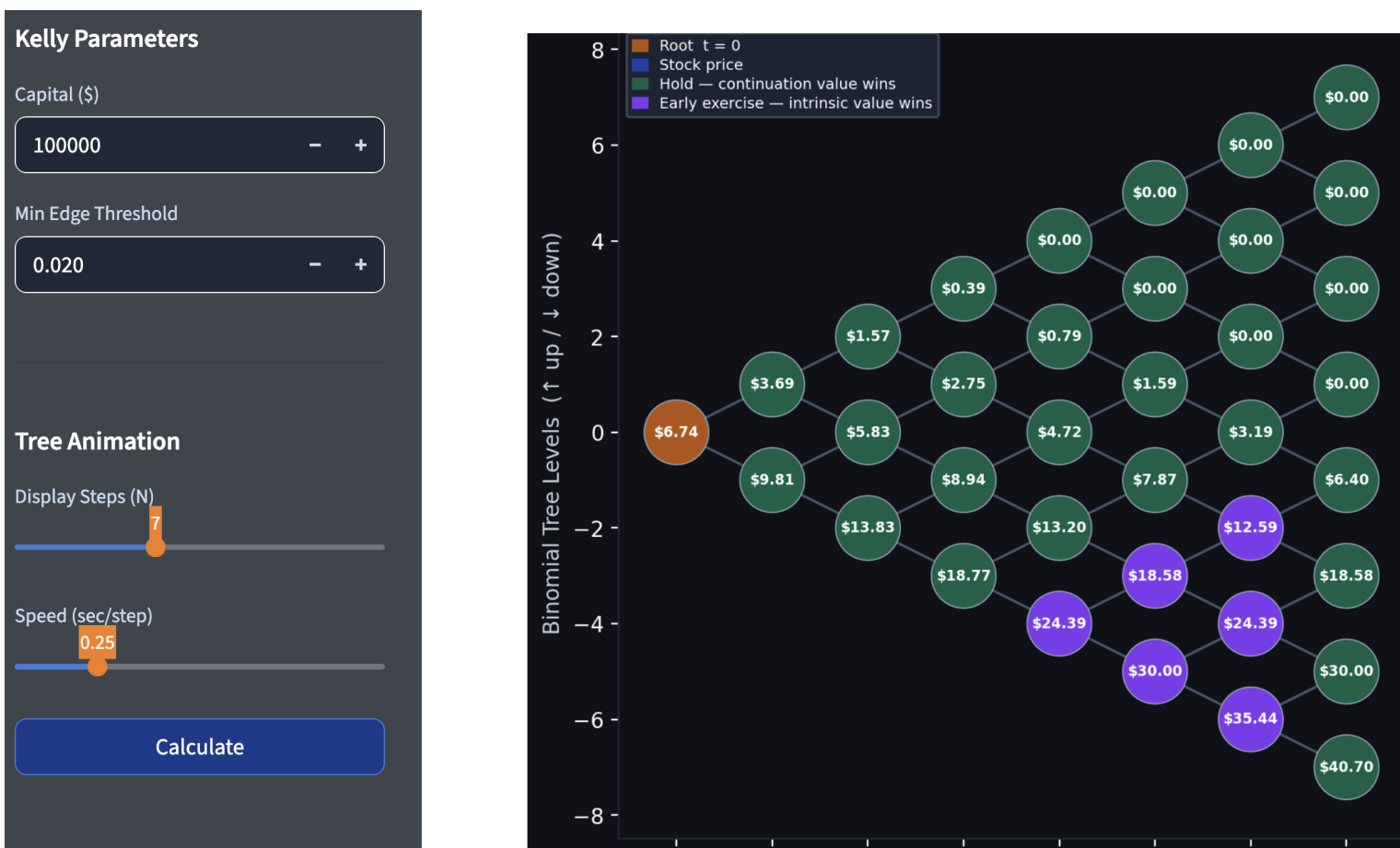
Forward + Backward Pass

Real AAPL put, 2020-03-16, herding regime ($S = \$241.47$, $K = \$240$, $DTE = 18\text{d}$).



Binomial Pricing Calculator

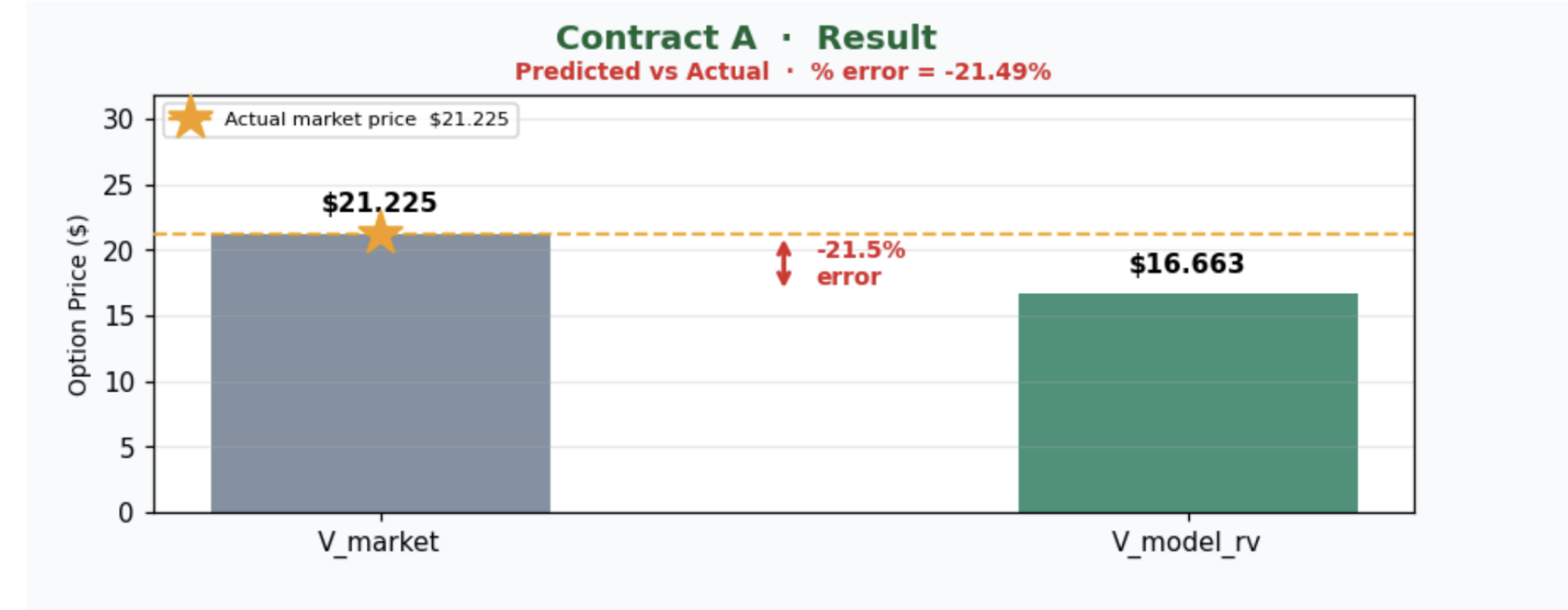
An interactive Streamlit application packages the full Layer-1 pricing pipeline behind a ticker-agnostic front end: enter S , K , DTE , r , σ , and the app renders the CRR tree node-by-node, the model fair value, the edge against an entered premium, the five Greeks, and the Kelly sizing recommendation.



Tree visualization. Each node shows $S(i, j) = S_0 \cdot u^{i-j} \cdot d^j$ from the forward pass; the interactive animation then runs backward induction with the American early-exercise check at every node, exposing the same diagnostic the research pipeline uses.

Results

Contract A (real AAPL put, herding regime): $S = \$241.47$, $K = \$240$, $DTE = 18\text{d}$. Both independent models arrive at the same trade decision:



Layer 1 (CRR) edge = -21.5% , Layer 2 (XGBoost) edge = -3.2% : **both layers say SELL** on this contract.

Performance summary (101,535 AAPL contracts, walk-forward):

Model	Brier	Hit rate	P&L (\$)
Baseline (random)	0.250	50.0%	0
Layer 1 CRR	—	63.0%	+1.4M
Layer 2 XGBoost	0.211	66.5%	+1.8M
L1 $\cap$ L2 agree (Q3)	—	76.5%	+2.9M

Cross-model agreement statistic: Pearson correlation coefficient $r = 0.413$ in herding regime ($n = 217$, $p = 2.4 \times 10^{-10}$); $r \approx 0$ in normal regime. Max drawdown $< 15\%$; **regime-conditioned** Kelly (half-Normal, quarter-Herding) keeps risk bounded.

Discussion & Conclusions

The central contribution is the *cross-validation result*: two **independent** models — one mathematical (CRR no-arbitrage), one empirical (XGBoost ML) — detect the same mispricing direction precisely in the regime behavioral finance theory predicts. Neither layer is trained on the other; concurrent signals therefore come from *shared underlying reality*.

Answers to the Introduction’s questions

- Yes** — CRR prices predict ITM/OTM. Layer-1 alone achieves a 63% per-trade hit rate on 101,535 AAPL contracts; Layer-2 (XGBoost) drives Brier score to 0.211, below the 0.25 naïve baseline.
- Yes** — cross-validation detects exploitable edge. Pooled $r = 0.032$ collapses to $r = 0.413$ ($p = 2.4 \times 10^{-10}$, $n = 217$) once the herding regime is isolated — a Simpson’s paradox in which both-SELL contracts (Q3) realize a 76.5% ITM rate vs. the $\sim 63\%$ baseline.
- Yes** — regime-conditioned Kelly converts edge into positive expected value. Walk-forward backtest produces **\\$2.94M** cumulative P&L on a $\$100\text{k}$ base across 3,760 executed trades (63.1% hit rate, max drawdown $< 15\%$, circuit breaker never fired).
- The framework’s most valuable output is the *no-trade* signal — when CRR and XGBoost disagree, no position is taken.

Limitations. AAPL 2016–2020 is a single equity in a single market cycle; CRR uses a flat IV per contract (no smile / skew); backtest assumes mid-price fills with no transaction friction; regime detection is backward-looking via IV/RV30.

Future work. Expand to SPY / QQQ / NDX options; integrate LSTM volatility forecaster; live paper-trading via IBKR / Alpaca; multi-leg spreads (vertical, calendar, iron condor) with joint Kelly sizing.

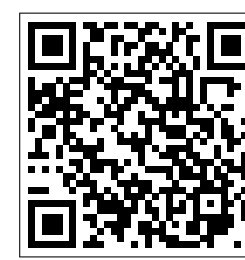
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Acknowledgments & Contact

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Code / Repository: github.com/dantzlerdc/CS495-Deep-Scholar



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